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MH3511 Data Analysis with Computer Group Project

What Makes a Player Valuable? The Role of Age, Performance, Playing Time, Club and Position

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Abstract:

The FIFA World cup 2022 reached record-breaking views of an estimated 5 billion people, making it the most watched sporting event in history and reinforcing football's status as the world's most popular sport. Beyond its global reach, football has evolved into a multi-billion-dollar industry where player market value plays a central role in shaping transfer decisions, club strategies and financial investments. Player market value serves as a key indicator of player's worth in the transfer market, influencing how clubs allocate resources and build competitive squads. However, determining a player's market value remains complex and multifaceted with indicators such as performance, club affiliation, age etc. We are going to study which factors strongly shape player valuation in an increasingly competitive global market.

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1. Introduction

In recent years, player market valuations have reached unprecedented levels. For instance, Alexander Isak was transferred to Liverpool for €125 million in the 2025-2026 season. With increasing commercialisation of the sport, clubs are spending more on high-profile players, increasing reliance on market valuations to assess player worth and allocate financial resources efficiently.

Our project uses a dataset containing the market value of Premier League players from the 2024-2025 season, including variables such as age, season statistics, club and position. We aim to answer the following questions:

1. Does the age of the player affect market valuation?
2. Does the number of goals and assists scored affect market valuation?
3. Does the number of matches won by a player affect market valuation?
4. Does the club a player is in affect market valuation?
5. Does the total minutes played in a season affect market valuation?
6. Does the position a player plays affect market valuation?

This report presents data description and analysis conducted using R programming and appropriate statistical data analysis techniques are applied to examine the relationships between market value and the variables. Results and findings are then interpreted with clear explanations and justifications provided to support conclusions drawn.

2. Data Description

The dataset “Football Data from Transfermarkt” is obtained from Kaggle, contains details of players, clubs, player market valuations and their attributes. The dataset consists of 10 csv data frames but only 4 are used for our project: “appearances.csv”, “players.csv”, “clubs.csv”, “clubs_games.csv”. Data originally taken from Transfermarkt, a football database that provides information on transfers, player performance and market values that is updated twice a year.

First, we performed a preliminary data cleaning to ensure that:

- Only 2024-25 season’s data are included in our dataset
- Only Premier league games are included in our dataset
- Redundant columns e.g. “date” and “game_id” are removed from dataset
- Player age computed using date of birth and end date of season to ensure consistency

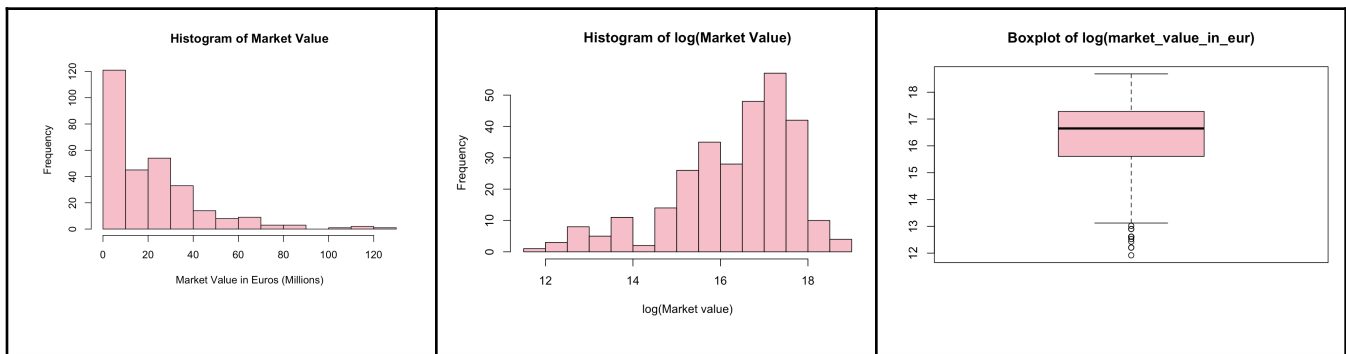
After cleaning the data, 294 observations (players) with 9 variables are retained for analysis:

1. `player_name`: name of player
2. `player_id`: unique id attached to player
3. `total_minutes_played`: total number of minutes played by a player across all matches in a season
4. `total_goals`: total number of goals scored in all matches in a season
5. `total_assists`: total number of assists in all matches in a season
6. `total_wins`: total number of matches won in a season
7. `sub_position`: position played by the player
8. `market_value_in_eur`: market value of a player for the 2024-25 season in euros
9. `age`: age of a player
10. `club`: Premier League club that a player is in during the 2024-25 season

3. Description and Cleaning of Dataset

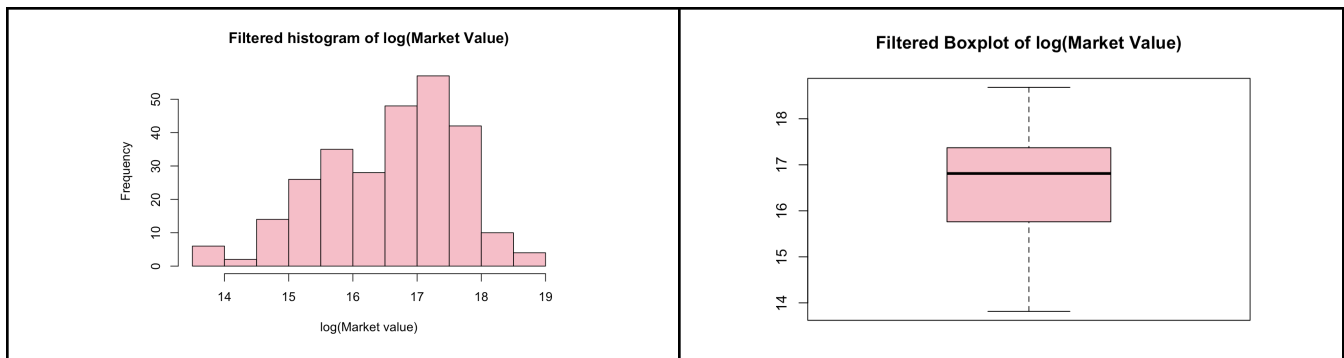
In this section, we analysed each variable individually to identify potential outliers and determine whether transformations are required to avoid highly skewed data.

3.1 Summary statistics for the main variable of interest, `market_value_in_eur`



We realised that the main variable, `market_value_in_eur`, was highly skewed. We applied a log transformation to the variable, and the log-transformed data contained outliers in the left tail. Therefore, we removed players with a market value below 1 million euros, representing approximately 7.5% of the data.

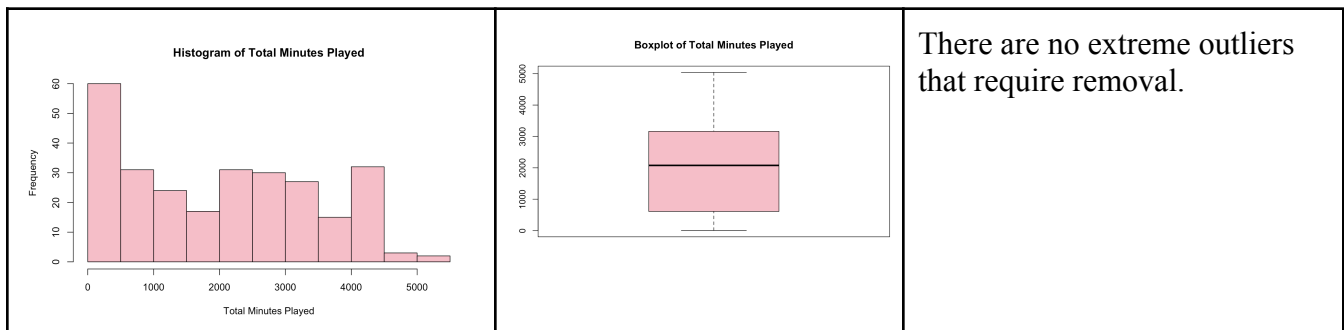
The resulting histogram and boxplot are shown below. The dataset is now more symmetric and does not have any outliers. We will proceed with this trimmed dataset.



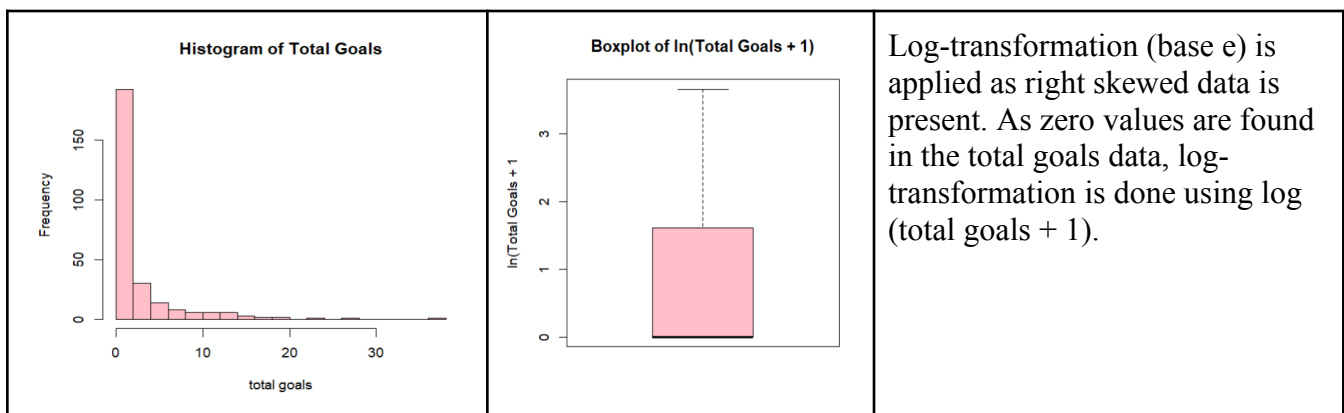
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
1.00	7.00	20.00	24.36	35.00	130.00

3.2 Summary statistics for other variables

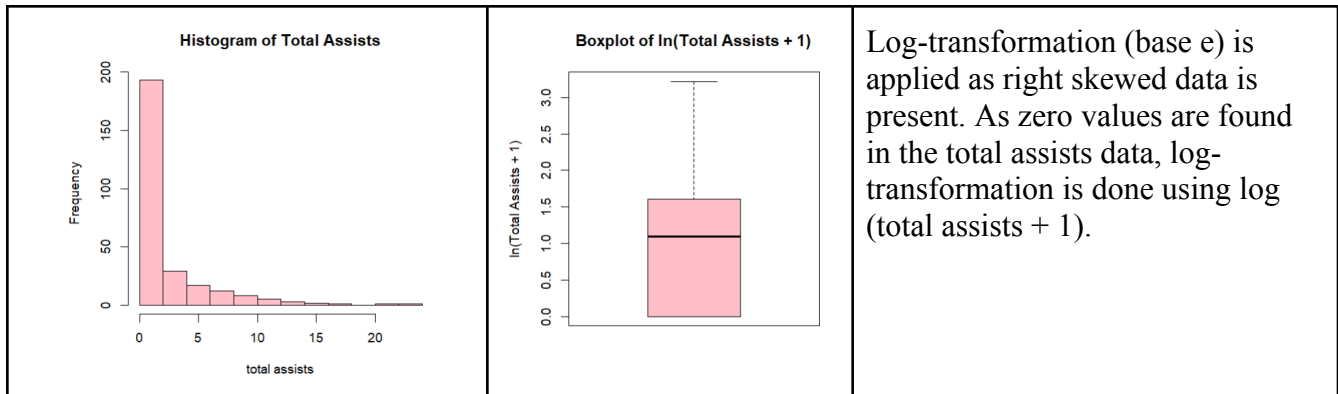
3.2.1 Total minutes played in a season, total_minutes_played



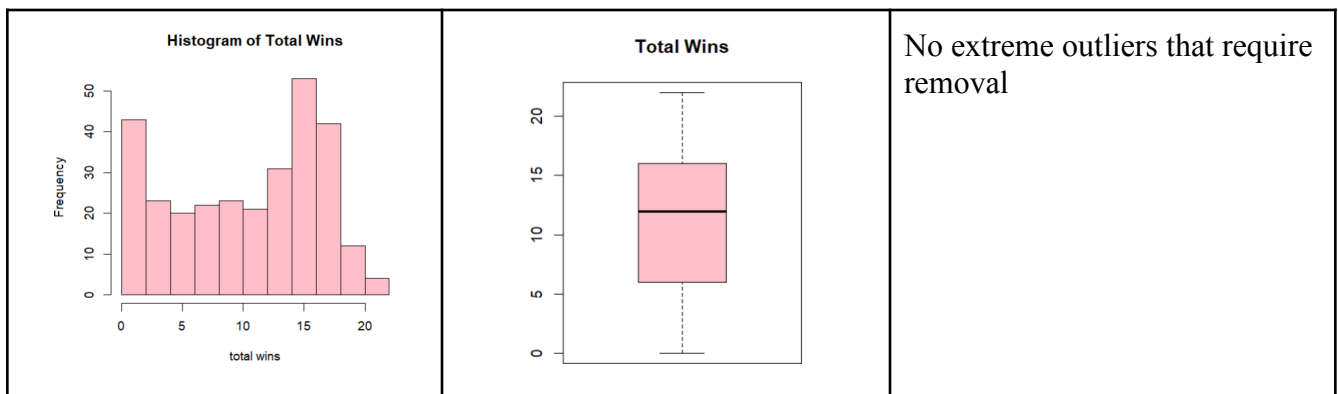
3.2.2 Total goals scored in a season, total_goals



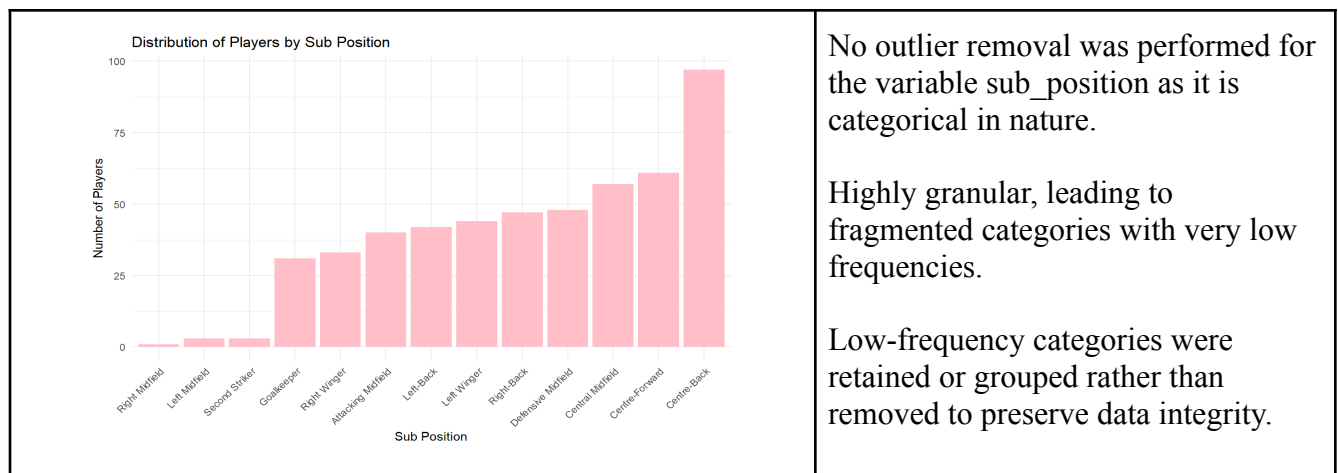
3.2.3 Total assists in a season, total_assists

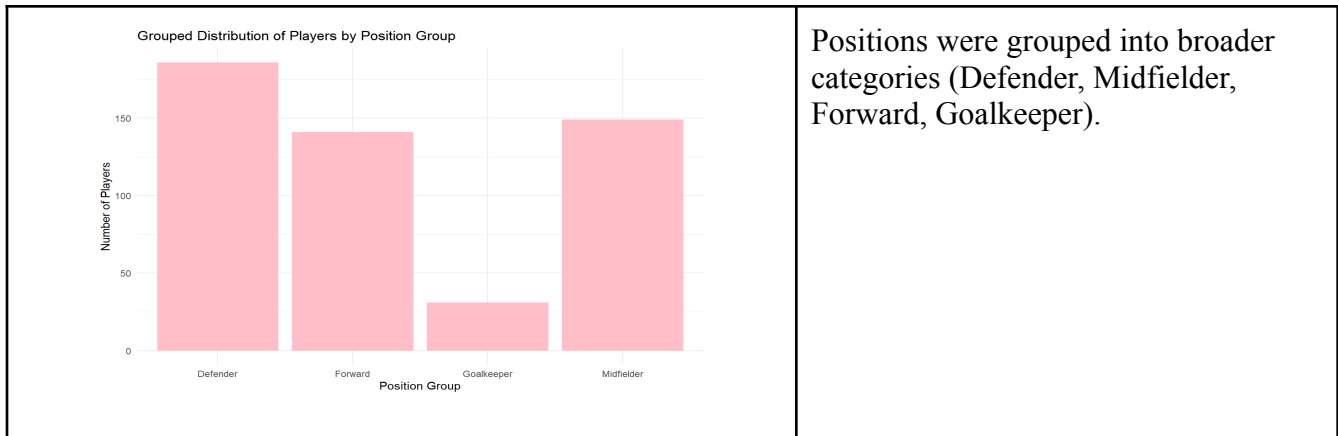


3.2.4 Total number of matches won in a season, total_wins

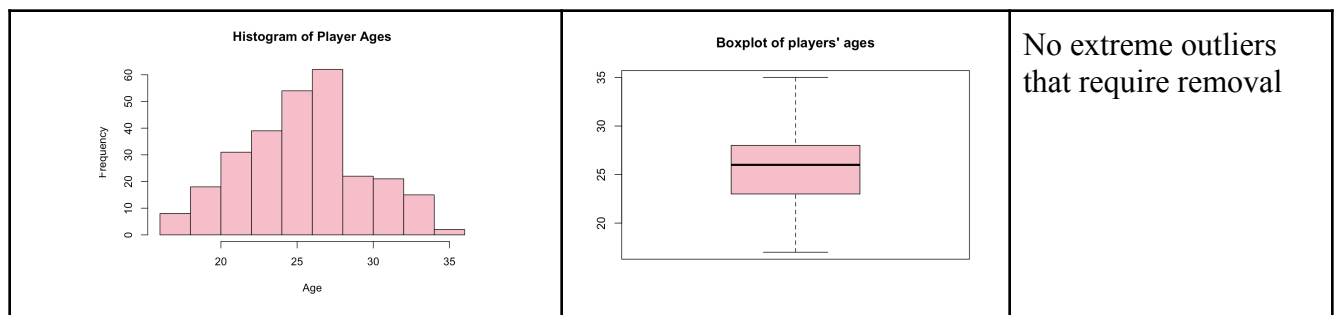


3.2.5 Position played by the player, sub_position

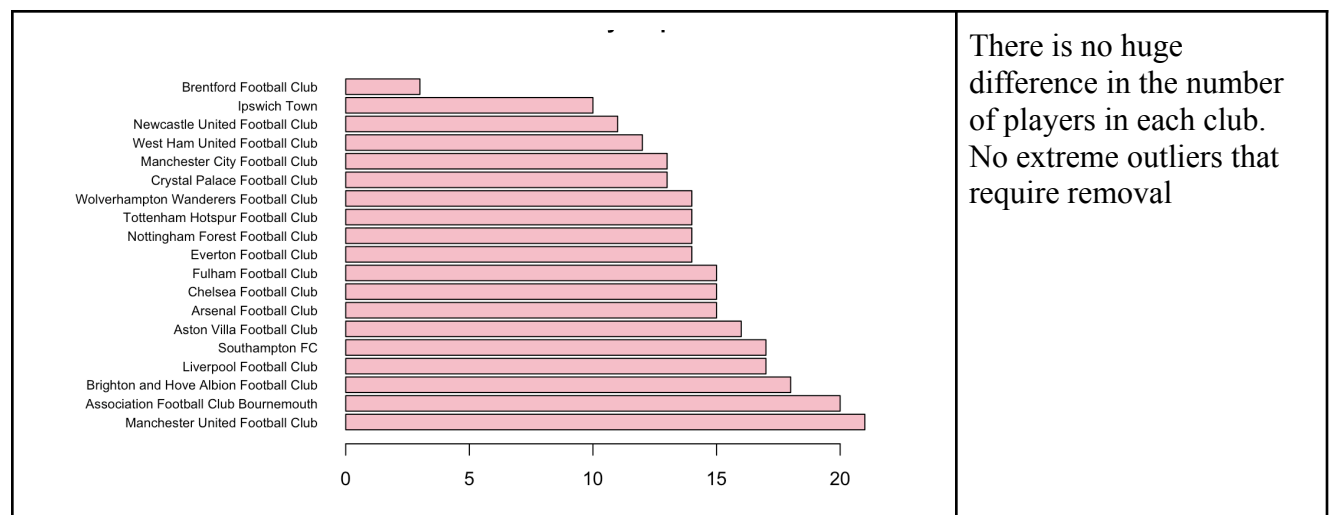




3.2.6 Age of the player, age



3.2.7 Club of the player, club



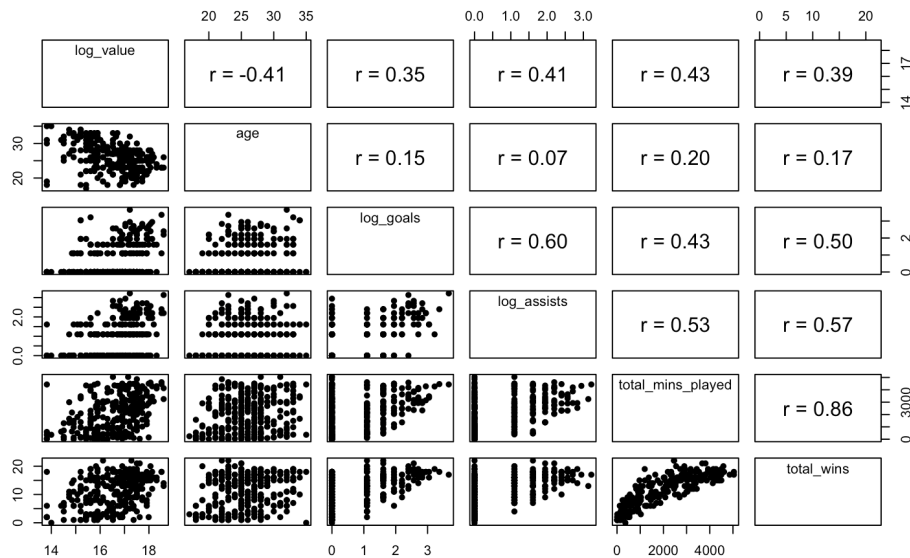
3.3 Final Dataset for Analysis

Based on the above analysis, the dataset remains at 272 observations with no further removal of outliers. Additionally, log-transformation has been applied to variables `market_value_in_eur`,

total_goals and total_assists to make skewed distributions more normal and reduce the impact of extreme outliers.

4. Statistical Analysis

4.1 Correlations between log(market_value_in_eur) and other variables



Scatter plots and correlation coefficients are useful in studying the possible linear relationships between a player's market value and the various continuous variables.

From the scatterplot matrix, it appears that log(market_value_in_eur) has moderate relationships with all the variables with age having a negative correlation while the rest of the variables have a positive correlation.

There are a few interesting observations from this tabulation:

- log_goals and log_assists have a strong relationship ($r = 0.60$)
- log_assists shows moderate relationships with total_minutes_played ($r = 0.53$) and total_wins ($r = 0.57$)
- total_minutes_played and total_wins exhibit a very strong relationship ($r = 0.86$)
- age shows very weak relationship with log_assists ($r = 0.07$)

We shall perform some statistical tests to confirm some of our observations in the next section.

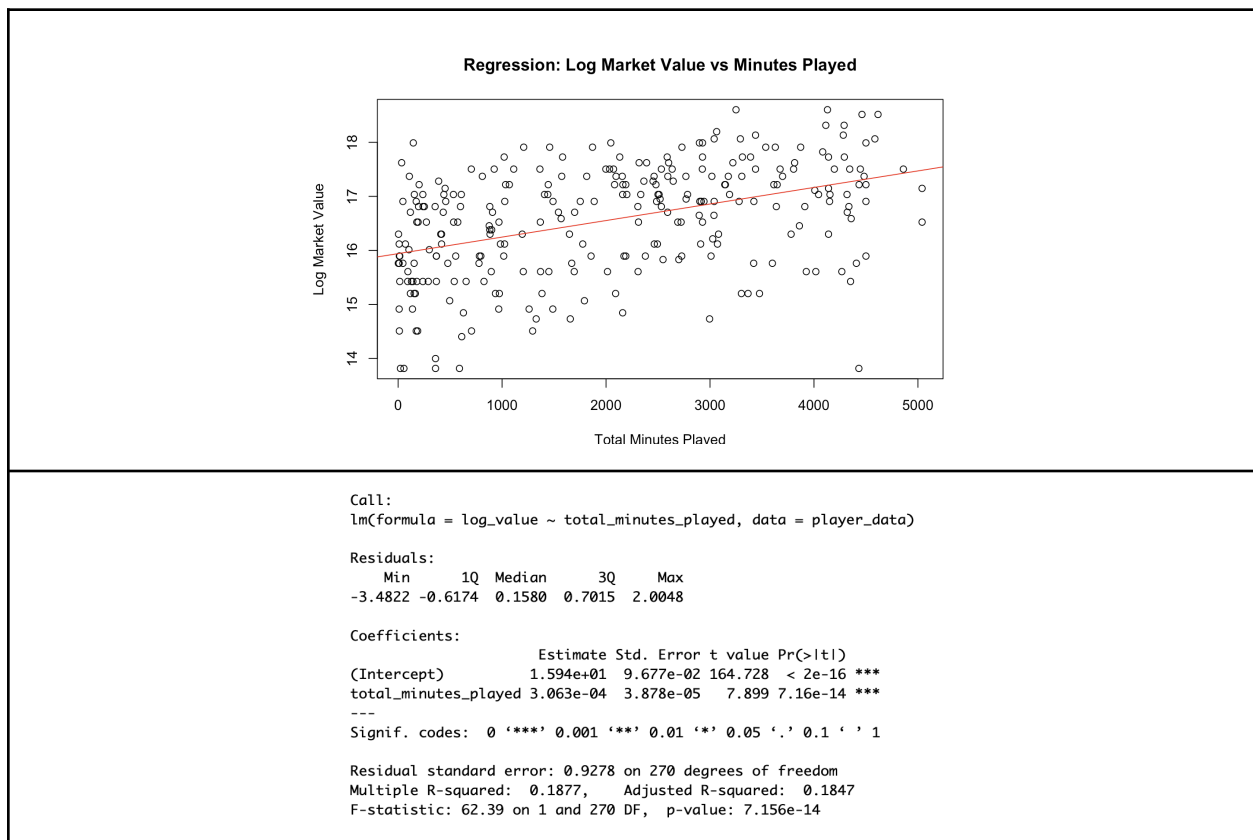
4.2 Statistical Tests

4.2.1 Relation between market value and total_minutes_played

In this section we try to answer the question “Does the total minutes played in a season affect market valuation?” A simple linear regression model is used to determine whether player market value depends on total_minutes_played.

The model is defined as $\log(\text{market_value_in_eur}) = \beta_0 + \beta_1(\text{total_minutes_played}) + \varepsilon$.

The following plot illustrates the scatter plot of $\log(\text{market_value_in_eur})$ against (total_minutes_played).



From the results, the regression model is $\log(\text{market_value_in_eur}) = 15.94 + 0.0003063 \times (\text{total_minutes_played})$.

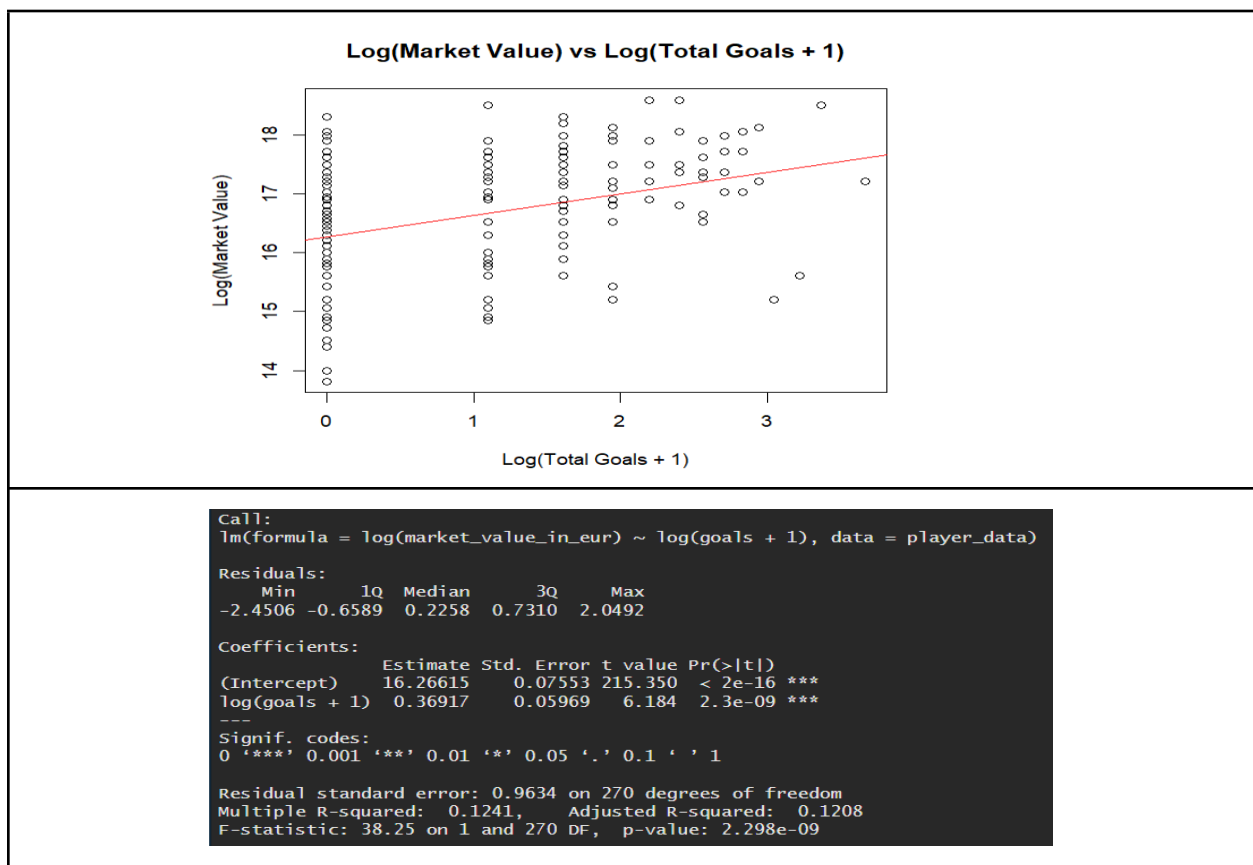
The coefficient for total_minutes_played is statistically significant (p-value = 7.16×10^{-14}), indicating that total_minutes_played has a significant effect on market value.

The R-squared value is 0.1877, which indicates that approximately 18.77% of the variation in $\log(\text{market_value_in_eur})$ can be explained by total_minutes_played. This suggests that while playing time has a noticeable influence on market value, a large proportion of the variation is still explained by other factors not included in the model.

4.2.2 Relation between market value and total_goals

In this section, we determine if the total number of goals scored affects the market value of the player by performing a simple linear regression between $\log(\text{market_value_in_eur})$ and $\log(\text{total_goals})$. Both variables have undergone a log-transformation to account for right-skewed data and zeros in goals, making sure that the data is symmetric.

At the 0.05 significance level, the regression model gives a p-value of 2.3×10^{-9} and slope 0.369. With a positive slope and small p-value (< 0.05), this suggests a strong positive correlation between players who score more goals and their market values. From the intercept value of 16.27, a player with zero goals is expected to have an estimated market value of $e^{16.266} = 1.14 \times 10^7$ euros (11.4 million euros). Even though the relationship is significant statistically, the multiple R-squared value of 0.1241 indicates that the total number of goals contribute to only about 12% of the variation in log market value. This implies that other factors such as total assists, sub position and total minutes played also impact the player's market value greatly.

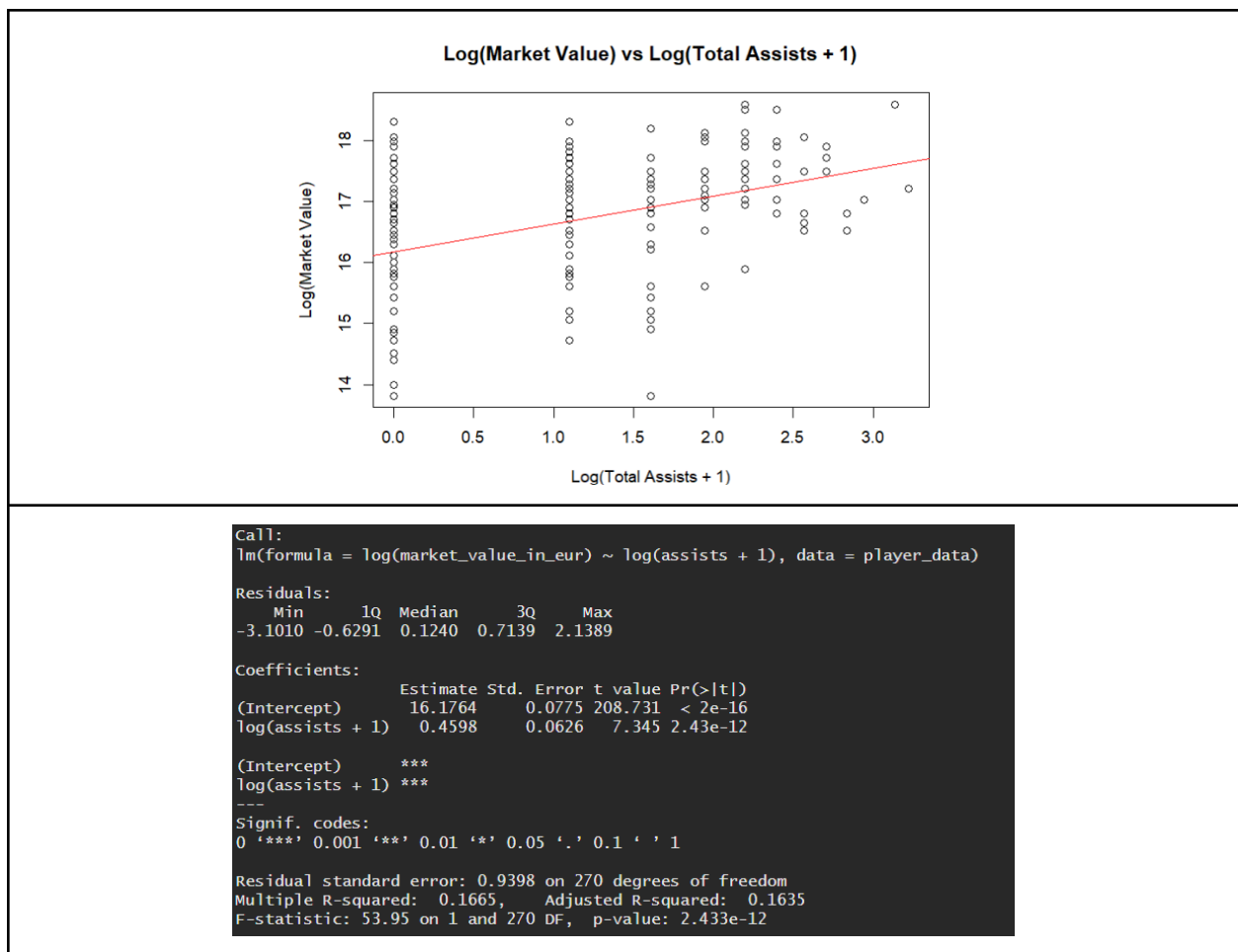


4.2.3 Relation between market value and total_assists

In this section, we want to determine if the total number of assists scored affects the market value of the player by performing a simple linear regression between $\log(\text{market_value_in_eur})$ and

$\log(\text{total_assists})$. Similar to the statistical analysis of the relation between market value and total goals, both variables had also undergone a log-transformation to account for right-skewed data and zeros in assists.

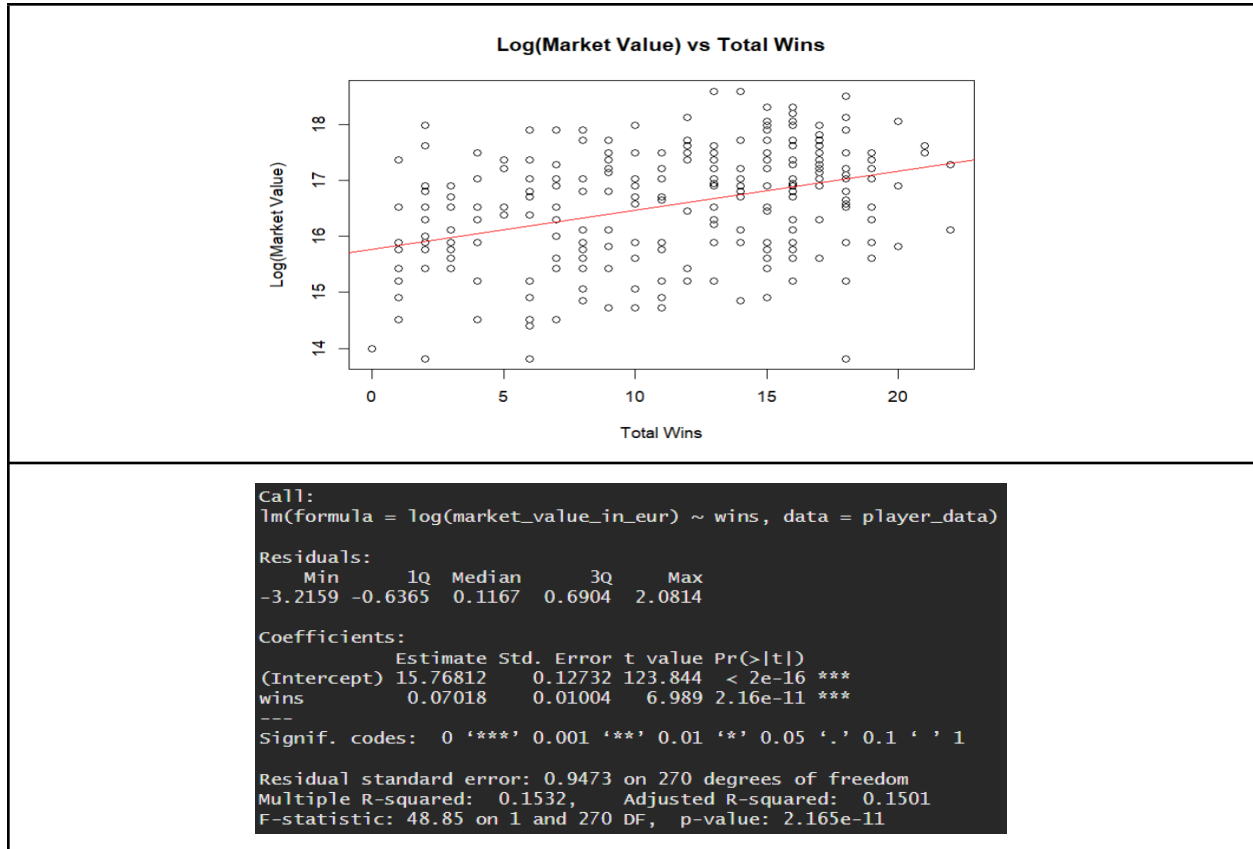
There is a strong positive correlation where players who score more assists will have higher market values. This can be seen from the regression model where at the 0.05 significance level, it gives a p-value of 2.43×10^{-12} and slope 0.460. From the intercept value of 16.18, a player with zero assists is predicted to have a market value of $e^{16.176} = 1.06 \times 10^7$ euros (10.6 million euros). Likewise, the multiple R-squared value of 0.1665 suggests that the total number of assists results in only about 16% of the variation in log market value, indicating that other factors like total goals, sub position and total minutes played also play an important role in determining market value.



4.2.4 Relation between market value and total_wins

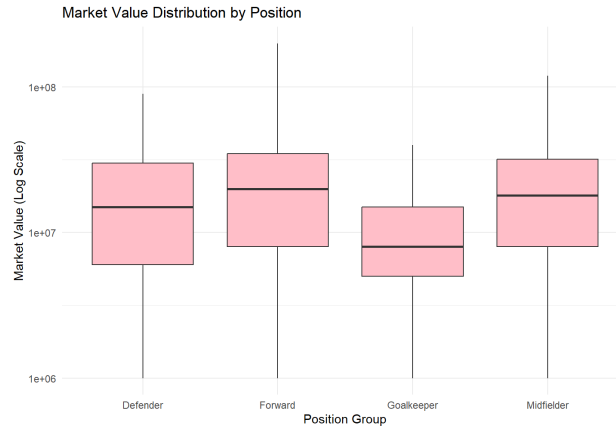
In this section, we attempted to answer the question “Does the number of matches won by a player affect market valuation?” This is done by performing a linear regression model between $\log(\text{market_value_in_eur})$ and total_wins .

At the 0.05 significance level, the regression model provides a p-value of 2.16×10^{-11} and slope 0.07. This indicates a strong positive correlation between the number of matches won by a player and their market values. From the intercept value of 15.8, a player who won zero matches is expected to have an estimated market value of $e^{15.77} = 7.06 \times 10^7$ euros (7.1 million euros). Even though the relationship is significant statistically, the multiple R-squared value of 0.1532 suggests that the total matches won contribute to only 15% of the variation in log market value. Similarly, it shows how there are also other influences that can affect the market value of a player.



4.2.5 Relation between market value and sub_position

To determine if a player's position affects their market valuation, we conducted a one-way ANOVA test to determine whether the mean of `log(market_value_in_eur)` differs across the categorical `sub_positions`. The plot below illustrates the distributions of `log(market_value_in_eur)` among the play positions:



The spreads of market value differ across positions, with forwards and midfielders showing greater variability compared to goalkeepers. Hence, the ANOVA test is appropriate for testing the inequality of the means. We test:

$$H_0: \mu_{\text{Defender}} = \mu_{\text{Forward}} = \mu_{\text{Goalkeeper}} = \mu_{\text{Midfielder}} \text{ against } H_1: \text{not all } \mu_i \text{ are equal}$$

```

              Df    Sum Sq   Mean Sq F value    Pr(>F)
position_group  3  7.105e+15  2.368e+15   4.887 0.00233 **
Residuals    503  2.437e+17  4.846e+14
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

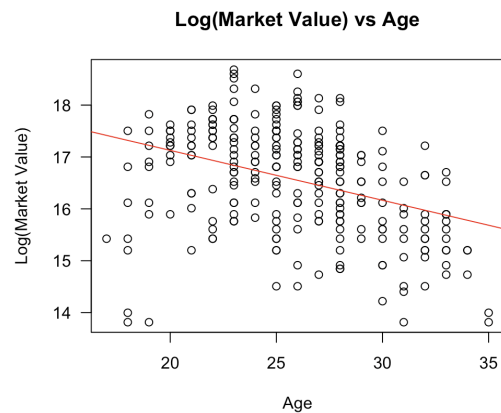
```

The ANOVA test returns a p-value of $0.00233 < 0.05$, which shows that the means are not equal at a significance level of 0.05. Therefore, we conclude that there is a statistically significant difference in market value across different positions.

4.2.6 Relation between market value and age

In this section, we determine whether the market value of a player depends on his age. This is done by performing a linear regression between $\log(\text{market_value_in_eur})$ and age.

The regression model provides a p-value of $4.11 \times 10^{-10} < 0.05$ and slope of -0.096, indicating a significant negative relationship between $\log(\text{market_value_in_eur})$ and age at 0.05 level of significance, and suggesting that older players tend to have lower market values. While the relationship is statistically significant, the multiple R-squared value of 0.1349 suggests that only ~13% of the variation in $\log(\text{market_value_in_eur})$ is explained by age, implying that other factors such as sub position and clubs are more likely to affect the market value of a player more greatly.



Call:
lm(formula = log(market_value_in_eur) ~ age, data = player_data)

Residuals:

Min	1Q	Median	3Q	Max
-3.5048	-0.6037	0.1645	0.6790	2.0525

Coefficients:

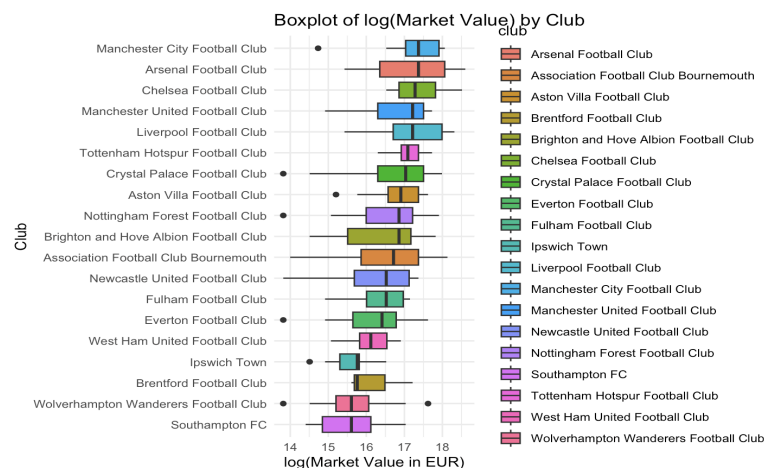
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	19.05237	0.38826	49.072	< 2e-16 ***
age	-0.09623	0.01483	-6.489	4.11e-10 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.967 on 270 degrees of freedom
Multiple R-squared: 0.1349, Adjusted R-squared: 0.1317
F-statistic: 42.11 on 1 and 270 DF, p-value: 4.11e-10

4.2.7 Relation between market value and clubs

In this section, we study whether the club a player belongs to affects his market valuation. The boxplot of `log(market_value_in_eur)` by club is first examined to compare the distributions visually. Top-tier clubs, such as Arsenal and Liverpool, tend to exhibit higher median log market values, suggesting a potential relationship between club affiliation and player market value.



We then test

$H_0: \mu_1 = \mu_2 = \dots = \mu_k$ against H_1 : not all club mean log market values are equal.

Since club is a categorical variable with multiple groups, a one-way ANOVA is used to test whether the mean log-transformed market value differs across clubs.

```
> summary(anova_club)
              Df Sum Sq Mean Sq F value    Pr(>F)
club           18  79.16   4.398    5.377 1.41e-10 ***
Residuals     253 206.93   0.818
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

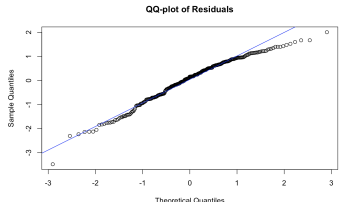
The ANOVA test returns a p-value of $1.41 \times 10^{-10} < 0.05$, we reject the null hypothesis at the 5% significance level. Therefore, we conclude that the mean log market value does differ significantly across clubs, indicating club affiliation plays an important role in influencing market valuation.

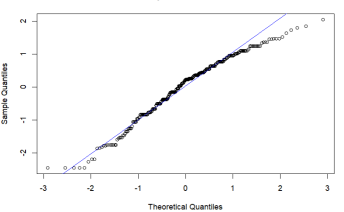
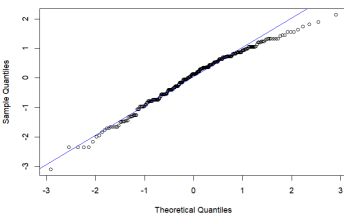
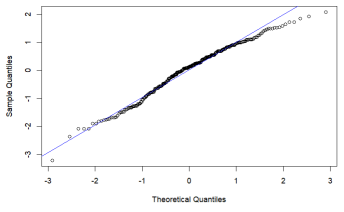
4.2.8 The single most important performance measure that is affecting market value?

As we summarise the relation between the performance measures and the player's market value from sections 4.1.1 to 4.1.4, we will then analyse the simple regression that was done to find out which of the 4 performance measures affect the market value of a player the most.

$$\log(\text{market_value_in_eur}) = \beta_0 + \beta_1 * X + \varepsilon$$

where X could be the variables total_minutes_played, log(total_goals+1), log(total_assists+1) and total_wins. From the p-values, R-squared values and residual qqplots shown in the summary table below, variable total_minutes_played is the single most important performance measure to model log(market_value_in_eur) using a simple linear model.

Variable(X)	Fitted Model with Y being log (market_value_in_eur)	p-value	R-square d	QQ-plot of residuals
total_minutes_played	$\hat{Y} = 15.94 + 0.0003063 \times X$	7.156×10^{-14}	0.1847	

Log (total_goals +1)	$\hat{Y} = 16.27 + 0.3692 \times X$	2.3×10^{-9}	0.1241	
log (total_assists +1)	$\hat{Y} = 16.18 + 0.4598 \times X$	2.43×10^{-12}	0.1665	
total_wins	$\hat{Y} = 15.68 + 0.0702 \times X$	2.16×10^{-11}	0.1532	

4.3 Multiple Linear Regression

In this section, we attempt to build a multiple linear model for $\log(\text{market_value_in_eur})$ based on the 4 given performance measures, $\text{total_minutes_played}$, $\log(\text{total_goals}+1)$, $\log(\text{total_assists}+1)$ and total_wins . We use a backward elimination method to select the most appropriate model.

Start: AIC=-53.46

```
log(market_value_in_eur) ~ total_minutes_played + log(total_goals +
1) + log(total_assists + 1) + total_wins
```

	Df	Sum of Sq	RSS	AIC
- total_wins	1	0.0477	215.45	-55.396
<none>			215.40	-53.456
- log(total_goals + 1)	1	3.1052	218.51	-51.563
- log(total_assists + 1)	1	5.8036	221.21	-48.224
- total_minutes_played	1	7.6945	223.10	-45.909

Step: AIC=-55.4

```
log(market_value_in_eur) ~ total_minutes_played + log(total_goals +
1) + log(total_assists + 1)
```

	Df	Sum of Sq	RSS	AIC
<none>			215.45	-55.396
- log(total_goals + 1)	1	3.0622	218.51	-53.557
- log(total_assists + 1)	1	5.7982	221.25	-50.172
- total_minutes_played	1	17.7882	233.24	-35.817

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.584254e+01	9.460089e-02	167.467091	3.111750e-273
total_minutes_played	2.109107e-04	4.483723e-05	4.703920	4.091659e-06
log(total_goals + 1)	1.372324e-01	7.031458e-02	1.951691	5.201637e-02
log(total_assists + 1)	2.156040e-01	8.028148e-02	2.685600	7.691783e-03

From the R output above, we conclude that total_minutes_played, log(total_goals+1) and log(total_assists+1) are significant variables that could be used to model the linear model log(market_value_in_eur), while total_wins is not a significant variable. The fitted model is:

$$\widehat{\log(\text{market_value_in_eur})} = 15.84 + 0.000211 * (\text{total_minutes_played}) + 0.137 * \log(\text{total_goals} + 1) + 0.216 * \log(\text{total_assists} + 1)$$

5. Conclusion

Football has evolved into a highly competitive and financially driven sport, where market valuation of a player is crucial in shaping transfer decisions and club strategies. This report examined how various factors affected the market valuation of Premier League players in the 2024-2025 season.

We concluded that:

- Age has a significant negative relationship with market value, indicating younger players tend to have higher valuations due to higher potential and longer career duration, but it does not affect market valuation as greatly as other variables.
- Player position and club affiliation significantly influence market value, highlighting the impact of position demand and club reputation on market valuation.
- Goals and assists are significant variables but total minutes played is the single most important performance predictor affecting player market value.
- Total wins are statistically insignificant when other performance variables are present, suggesting that individual contributions outweigh team results when determining player market value.

However, there are several limitations. Firstly, our analysis is based on a single season (2024-2025), which may fail to capture long-term trends or fluctuations in players' performance and valuation. Secondly, other important factors such as injuries, contract length, market demand and club financial strength were not considered. Future research could incorporate machine learning approaches and include additional variables to accurately capture the complexity of player valuation.

In conclusion, this study provides empirical evidence that (performance and playing time) are important determinants of player market value, offering practical insights for clubs and stakeholders in evaluating talent.

6. Appendix

Preliminary Data Cleaning

```
library(tidyquant)
library(tidyverse)
library(lubridate)

appearances <- read.csv("appearances.csv", header = TRUE)
start_date <- as.Date("2024-08-16")
end_date <- as.Date("2025-05-25")

#filter to premier league appearance (GB1)
appearances2 <- appearances[appearances$date >= start_date & appearances$date <= end_date,]
appearances2 <- appearances2 %>%
  filter(competition_id == "GB1") %>%
  select(player_id, game_id, player_club_id, date, player_name, competition_id, minutes_played, goals, assists)

#merge dataset to get player details, include their age and relevant columns
players <- read.csv("players.csv", header = TRUE)
players2 <- merge(appearances2, players, by = "player_id")
players2 <- players2 %>%
  mutate(age = floor(time_length(interval(players2$date_of_birth, "2025-05-25"), "years"))) %>%
  select(player_id, sub_position, market_value_in_eur, player_name, player_club_id, game_id, minutes_played, goals, assists, age)

clubs <- read.csv("clubs.csv", header = TRUE)
colnames(clubs)[1] <- "player_club_id"
clubs <- select(clubs, player_club_id, name)
View(clubs)
players2 <- merge(clubs, players2, by = "player_club_id")
View(players2)

club_wins <- read.csv("club_games.csv", header = TRUE)
club_wins2 <- merge(club_wins, players2, by = "game_id") %>%
  select(name, player_id, sub_position, market_value_in_eur, player_name, minutes_played, goals, assists, is_win, age)
View(club_wins2)

player_data <- club_wins2 %>%
  group_by(player_name) %>%
  summarise(player_id = first(player_id),
    total_minutes_played = sum(minutes_played),
    total_goals = sum(goals),
    total_assists = sum(assists),
    total_wins = sum(is_win),
    sub_position = first(sub_position),
    market_value_in_eur = first(market_value_in_eur),
    age = first(age),
    club = first(name),
    .groups = "drop"
  ) %>%
  filter(market_value_in_eur >= 1000000)
View(player_data) #final dataset to be used
```

Output of player_data (final dataset used with 272 entries):

	player_name	player_id	total_minutes_played	total_goals	total_assists	total_wins	sub_position	market_value_in_eur	age	club
1	Aaron Ramsdale	427568	3780	0	0	17	Goalkeeper	1.2e+07	27	Southampton FC
2	Aaron Wan-Bissaka	477758	4150	4	6	17	Right-Back	2.2e+07	27	West Ham United Football Club
3	Abdoulaye Doucouré	127187	4018	4	2	16	Attacking Midfield	6.0e+06	32	Everton Football Club
4	Abdukodir Khusanov	763079	108	0	0	1	Centre-Back	3.5e+07	21	Manchester City Football Club
5	Adam Armstrong	250426	2378	4	2	16	Centre-Forward	8.0e+06	28	Southampton FC
6	Adam Wharton	744149	1870	0	2	7	Defensive Midfield	6.0e+07	21	Crystal Palace Football Club
7	Alejandro Garnacho	811779	2926	8	4	17	Left Winger	4.0e+07	20	Manchester United Football Club
8	Alex Iwobi	242631	4044	14	4	18	Left Winger	2.5e+07	29	Fulham Football Club
9	Alex Scott	855256	924	0	0	10	Central Midfield	4.0e+07	21	Association Football Club Bournemouth
10	Alexis Mac Allister	534033	3064	4	4	16	Central Midfield	8.0e+07	26	Liverpool Football Club
11	Alisson	105470	3038	0	0	11	Goalkeeper	1.7e+07	32	Liverpool Football Club
12	Amad Diallo	536835	2928	16	14	17	Right Winger	5.0e+07	22	Manchester United Football Club
13	Amadou Onana	485706	2602	4	0	16	Defensive Midfield	4.5e+07	23	Aston Villa Football Club
14	Andreas Pereira	203394	2690	4	6	15	Attacking Midfield	1.5e+07	29	Fulham Football Club
15	André	800176	2784	0	0	17	Defensive Midfield	2.5e+07	23	Wolverhampton Wanderers Football Club
16	André Onana	234509	4140	0	0	16	Goalkeeper	1.2e+07	29	Manchester United Football Club
17	Andrés García	1063329	10	0	0	1	Right-Back	7.0e+06	22	Aston Villa Football Club
18	Anthony Elanga	583189	3438	10	12	19	Right Winger	4.0e+07	23	Nottingham Forest Football Club
19	Antoine Semenyo	583255	4282	18	8	18	Right Winger	7.5e+07	25	Association Football Club Bournemouth
20	Antonee Robinson	349701	4318	0	18	17	Left-Back	2.5e+07	27	Fulham Football Club
21	Antony	602105	202	0	0	5	Right Winger	3.0e+07	25	Manchester United Football Club

Showing 1 to 21 of 272 entries, 10 total columns

Data Cleaning (Removing Outliers and Log Transformation)

Total Goals, Total Assists, Total Wins

```
goals <- player_data$total_goals
ln_goals <- log(goals)
hist(goals,
     main="Histogram of Total Goals",
     xlab = "total goals", col="pink", breaks=15)
boxplot(log(goals+1),
        col="pink", main="Boxplot of ln(Total Goals + 1)",
        ylab = "ln(Total Goals + 1)")

assists <- player_data$total_assists
hist(assists,
     main="Histogram of Total Assists",
     xlab = "total assists", col="pink", breaks=10)
boxplot(assists, col="pink", main="Total Assists")
summary(assists)
boxplot(log(assists + 1),
        col="pink", main="Boxplot of ln(Total Assists + 1)",
        ylab = "ln(Total Assists + 1)")

wins <- player_data$total_wins
hist(wins,
     main="Histogram of Total Wins",
     xlab = "total wins", col="pink", breaks=10)
boxplot(wins, col="pink", main="Total Wins")
```

Total Minutes Played

```
hist(player_data$total_minutes_played,
     main = "Histogram of Total Minutes Played",
     xlab = "Total Minutes Played",
     col = "pink")

boxplot(player_data$total_minutes_played,
        main = "Boxplot of Total Minutes Played", col = "pink")

summary(player_data$total_minutes_played)
```

Sub Position

```
#plotting the data
ggplot(player_data, aes(x = reorder(sub_position, sub_position, length))) +
  geom_bar(fill = "pink") +
  labs(
    title = "Distribution of Players by Sub Position",
    x = "Sub Position",
    y = "Number of Players"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

#group by the position
player_data <- player_data %>%
  mutate(position_group = case_when(
    sub_position %in% c("Centre-Back", "Left-Back", "Right-Back") ~ "Defender",
    sub_position %in% c("Defensive Midfield", "Central Midfield", "Attacking Midfield",
      "Left Midfield", "Right Midfield") ~ "Midfielder",
    sub_position %in% c("Centre-Forward", "Second Striker", "Left Winger", "Right Winger") ~ "Forward",
    sub_position %in% c("Goalkeeper") ~ "Goalkeeper",
    TRUE ~ "Other"
  ))

#clearer plot
ggplot(player_data, aes(x = position_group)) +
  geom_bar(fill = "pink") +
  labs(
    title = "Grouped Distribution of Players by Position Group",
    x = "Position Group",
    y = "Number of Players"
  ) +
  theme_minimal()
```

Age

```
hist(player_data$age, main = "Histogram of Player Ages", xlab = "Age",
  col = "pink")
boxplot(player_data$age, main="Boxplot of players' ages", col="pink")
```

Club

```
#barplot
par(mar = c(5, 12, 4, 2))

club_freq <- table(player_data$club)
club_freq <- sort(club_freq, decreasing = TRUE)

barplot(
  club_freq,
  horiz = TRUE,
  las = 1,
  col = "pink",
  main = "Number of Players per Club",
  xlab = "Frequency",
  cex.names = 0.7
)

player_data <- player_data %>%
  mutate(log_market_value = log(market_value_in_eur))
```

Statistical Analysis

Correlation between `log(market_value_in_eur)` and other variables

```
player_data$market_value_in_eur <- as.numeric(player_data$market_value_in_eur)
player_data$age <- as.numeric(player_data$age)
player_data$total_goals <- as.numeric(player_data$total_goals)
player_data$total_assists <- as.numeric(player_data$total_assists)
player_data$total_minutes_played <- as.numeric(player_data$total_minutes_played)
player_data$total_wins <- as.numeric(player_data$total_wins)
```

```
player_data$log_value <- log(player_data$market_value_in_eur)
player_data$log_goals <- log(player_data$total_goals + 1)
player_data$log_assists <- log(player_data$total_assists + 1)
```

```
X <- data.frame(
  log_value = player_data$log_value,
  age = player_data$age,
  log_goals = player_data$log_goals,
  log_assists = player_data$log_assists,
  total_mins_played = player_data$total_minutes_played,
  total_wins = player_data$total_wins
)
```

```
panel.cor <- function(x, y, digits = 2, prefix = "r = ", cex.cor = 1.5) {
  usr <- par("usr")
  on.exit(par(usr))
  par(usr = c(0, 1, 0, 1))
  r <- cor(x, y, use = "complete.obs")
  txt <- paste0(prefix, format(round(r, digits), nsmall = digits))
  text(0.5, 0.5, txt, cex = cex.cor)
}
```

```
panel.scatter <- function(x, y, ...) {
  points(x, y, pch = 16)
}
```

```
panel.empty <- function(x, ...) {
  usr <- par("usr")
  on.exit(par(usr))
  par(usr = c(0, 1, 0, 1))
}

pairs(
  X,
  upper.panel = panel.cor,
  lower.panel = panel.scatter,
  diag.panel = panel.empty
)
```

Statistical tests:

Total Goals, Total Assists, Total Wins

```
model1 <- lm(log(market_value_in_eur) ~ log(goals + 1), data = player_data)
summary(model1)
plot(log(goals + 1), log(player_data$market_value_in_eur),
     xlab = "Log(Total Goals + 1)",
     ylab = "Log(Market Value)",
     main = "Log(Market Value) vs Log(Total Goals + 1)")
abline(model1, col="red")

model2 <- lm(log(market_value_in_eur) ~ log(assists + 1), data = player_data)
summary(model2)
plot(log(assists + 1), log(player_data$market_value_in_eur),
     xlab = "Log(Total Assists + 1)",
     ylab = "Log(Market Value)",
     main = "Log(Market Value) vs Log(Total Assists + 1)")
abline(model2, col="red")

model3 <- lm(log(market_value_in_eur) ~ wins, data = player_data)
summary(model3)
plot(wins, log(player_data$market_value_in_eur),
     xlab = "Total Wins",
     ylab = "Log(Market Value)",
     main = "Log(Market Value) vs Total Wins")
abline(model3, col="red")

qqnorm(residuals(model1), main="QQ-plot of Residuals")
qqline(residuals(model1), col="blue")

qqnorm(residuals(model2), main="QQ-plot of Residuals")
qqline(residuals(model2), col="blue")

qqnorm(residuals(model3), main="QQ-plot of Residuals")
qqline(residuals(model3), col="blue")
```

Total Minutes Played

```
model_minutes <- lm(log_value ~ total_minutes_played, data = player_data)
summary(model_minutes)

plot(player_data$total_minutes_played, player_data$log_value,
     xlab = "Total Minutes Played",
     ylab = "Log Market Value",
     main = "Regression: Log Market Value vs Minutes Played")

abline(model_minutes, col = "red")

qqnorm(residuals(model_minutes), main="QQ-plot of Residuals")
qqline(residuals(model_minutes), col="blue")
```

Sub Position

```
#Analysis of data
market_value_in_eur <- log(player_data$market_value_in_eur)
anova_model <- aov(market_value_in_eur ~ position_group, data = player_data)
summary(anova_model)

ggplot(player_data, aes(x = position_group, y = market_value_in_eur)) +
  geom_boxplot(fill = "pink") +
  scale_y_log10() +
  labs(
    title = "Market Value Distribution by Position",
    x = "Position Group",
    y = "Market Value (Log Scale)"
  ) +
  theme_minimal()
```

Age

```
plot(player_data$age, log(player_data$market_value_in_eur),
     main = "Log(Market Value) vs Age",
     xlab = "Age",
     ylab = "Log(Market Value)",
     pch = 1,
     las = 1)
abline(model_age, col = "red")
```

Club

```
#boxplot
ggplot(player_data, aes(x = reorder(club, log_market_value, FUN = median),
                        y = log_market_value, fill = club)) +
  geom_boxplot() +
  coord_flip() +
  labs(
    title = "Boxplot of log(Market Value) by Club",
    x = "Club",
    y = "log(Market Value in EUR)"
  ) +
  theme_minimal()

#anova test
anova_club <- aov(log_market_value ~ club, data = player_data)
summary(anova_club)
```

Multiple Linear Regression

```
full_model <- lm(log(market_value_in_eur) ~ total_minutes_played +  
                log(total_goals + 1) +  
                log(total_assists + 1) +  
                total_wins,  
                data = player_data)  
step_model <- step(full_model, direction = "backward")  
summary(step_model)$coefficients
```

7. References

David Cariboo. (n.d.). *Football Data from Transfermarkt* [Dataset]. Kaggle.

<https://www.kaggle.com/datasets/davidcariboo/player-scores>

(dataset is updated every week, hence dataset that was used in our project is in

<https://github.com/hh44111/mh3511-assignment/tree/main> under csv files.zip)